

CSE655
Network Science
Winter 2026

Mini Project Report

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A Network Analysis of Anime Viewing Behaviour

Abstract

This project analyzes large-scale anime viewing behavior using network science techniques. By constructing a co-watch network from user interaction data, we uncover structural properties, identify critical anime, and study robustness and evolution. The results show that the anime ecosystem is highly interconnected, exhibits small-world properties, and is driven by a few structurally important hub anime.

1. Introduction

Understanding how users consume anime provides insights into content similarity, recommendation systems, and cultural trends. Instead of relying on metadata, we model relationships based on real user behavior such as watching, dropping, and wishlisting anime.

The primary goal of this project is to:

- Construct an anime co-watch network
- Analyze structural properties of the network
- Identify central and critical anime
- Study robustness under failures
- Examine temporal evolution of the network

2. Dataset

We use the MyAnimeList dataset containing:

- ~109 million user-anime interactions
- ~17,000 anime titles
- ~300,000 users

Due to computational constraints, we perform sampling:

- Selected users using: $\text{user_id} \% 10 == 0$
- Preserves diversity and avoids bias
- Final dataset:
 - ~32,000 users
 - ~17,500 anime

3. Network Construction

We construct a co-watch network where:

- Nodes represent anime
- An edge exists if a minimum number of users have watched both anime

Threshold used:

- Minimum co-watch count: 2000 users. This threshold filters out weak relationships and emphasizes strong co-viewing patterns.

Resulting network:

- Nodes: ~1000
- Edges: ~43,000
- Density: ~0.087

Users in this set: 31,151

Anime in this set: 1,000

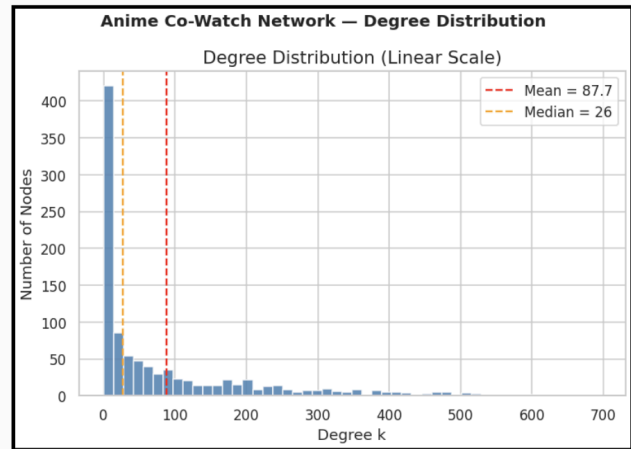
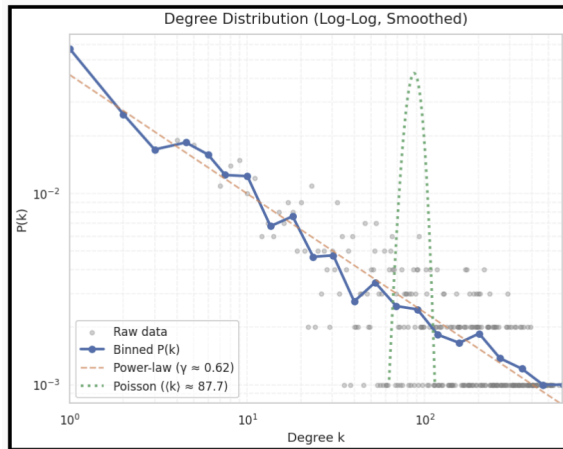
Co-watch network built:

Nodes (anime): 1,000

Edges: 43,844

Density: 0.08778

4. Degree Distribution

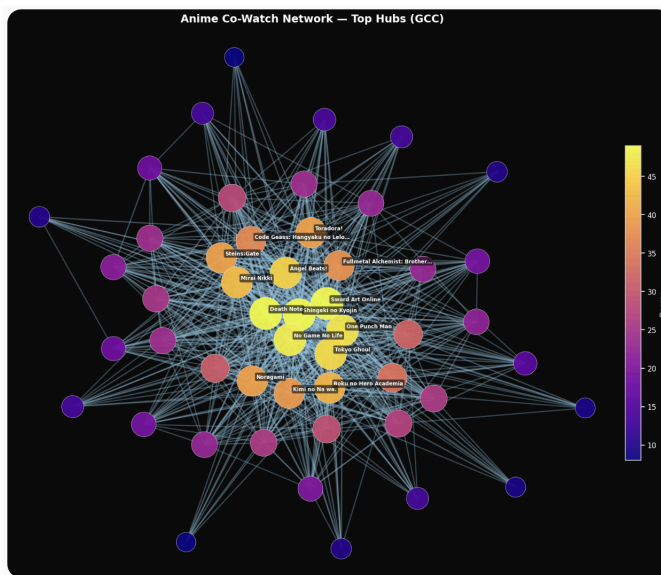


The degree distribution shows:

- Heavy-tailed behavior
- Not a pure power-law
- Not random (non-Poisson)

Interpretation:

- Few anime act as hubs with very high connectivity
- Most anime have low degree

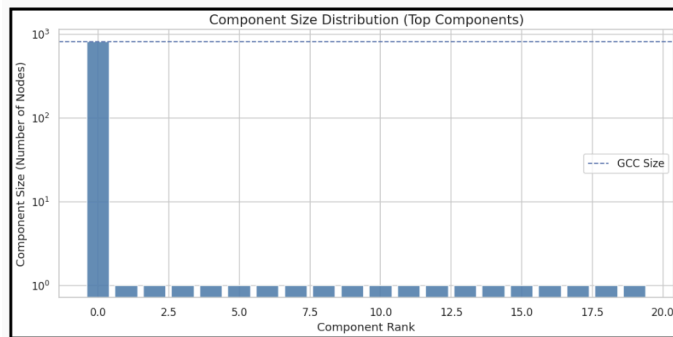


Anime Co-Watch Network Analysis: This graph focuses on the top 50 most connected anime after filtering out weaker links. We can see the core part of the network, where popular anime are tightly connected because a lot of people tend to watch them together.

The filtered visualization of the top connected anime highlights a dense core where popular titles are strongly interconnected due to widespread co-viewing.

Conclusion: The network is heterogeneous but not strictly scale-free

5. Structural Properties



=== Connected Components Analysis ===

Total nodes: 1,000
Total edges: 43,844
Total components: 193

--- Giant Connected Component (GCC) ---

GCC size (nodes): 808 (80.80%)
GCC edges: 43,844
GCC density: 0.134479
2nd largest component: 1 nodes
Isolated nodes: 192 (19.20%)

5.1 Giant Connected Component (GCC)

- ~80% of nodes belong to GCC; Indicates strong interconnectivity

Co-Watch Network vs ER Random Graph		
Metric	Observed	ER Random
Nodes	1,000	1,000
Edges	43,844	43,853
Density	0.08778	0.08779
GCC fraction	0.8080	1.0000
Avg clustering	0.6642	0.0881
Transitivity	0.5302	0.0881
Avg path length	1.980	1.544

5.2 Small-World Property

Compared to random graphs:

- High clustering coefficient; Short average path length

The anime network exhibits small-world properties, implying that users tend to watch clusters of related anime, while still remaining globally connected. This structure enables efficient information flow, which is beneficial for recommendation systems.

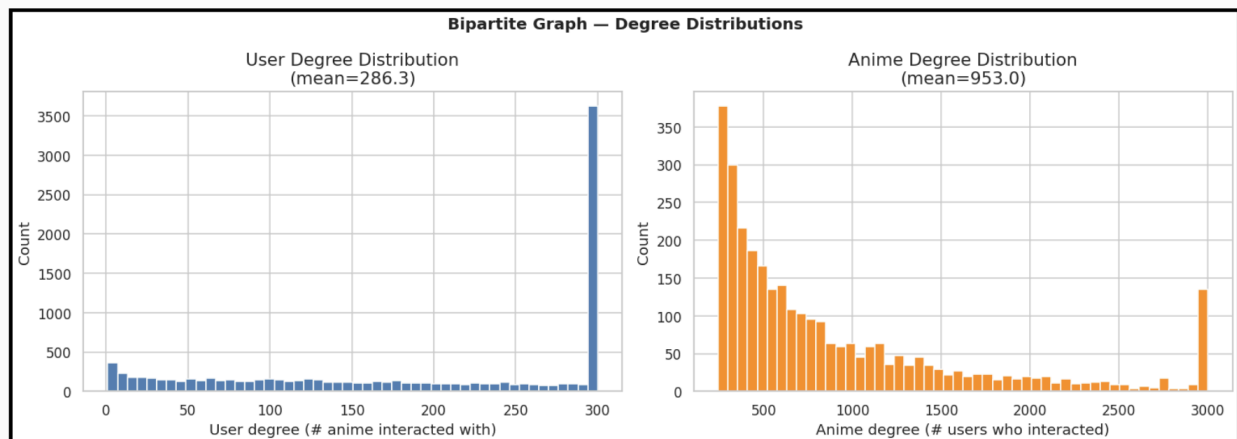
6. Bipartite Network and Projection

We model the system as a bipartite graph:

- Users $\leftrightarrow$ Anime

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Bipartite graph built:  
User nodes:      9,987  
Anime nodes:     3,000  
Edges:           2,859,083  
Is bipartite: True
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Number of connected components: 1
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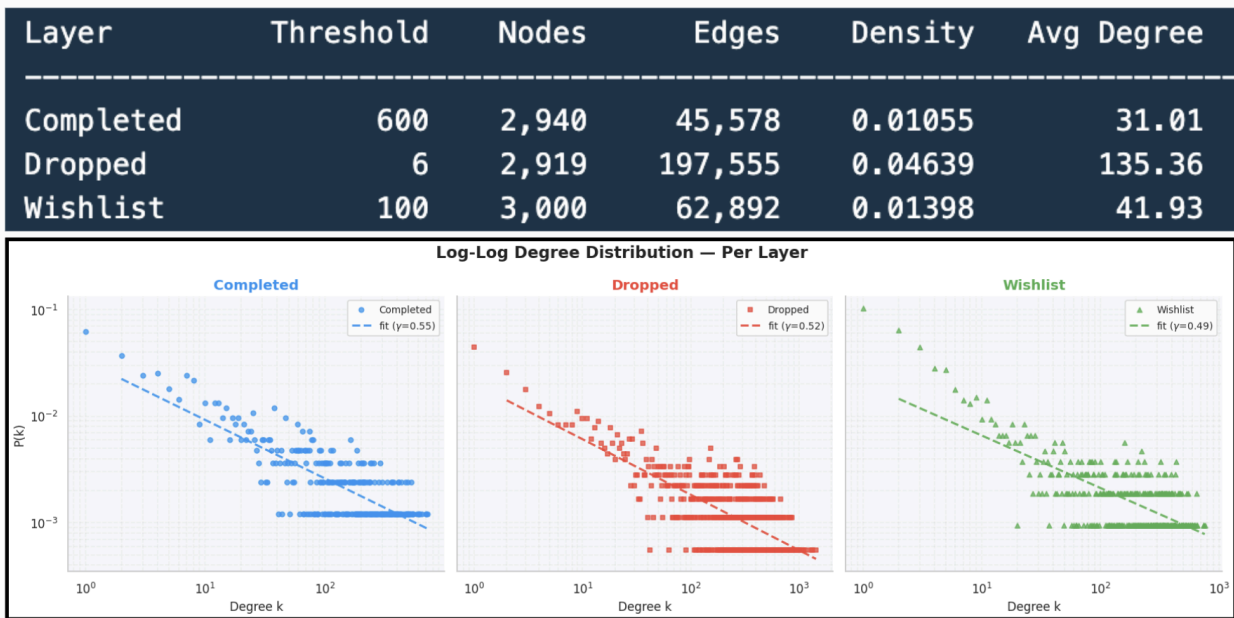
We project onto anime layer for:

- Completed
- Dropped
- Wishlist

Insights:

- **Completed:** Reflects similarity in viewing behavior
- **Dropped:** Captures shared dissatisfaction or disengagement
- **Wishlist:** Indicates future interest and aspirational viewing patterns

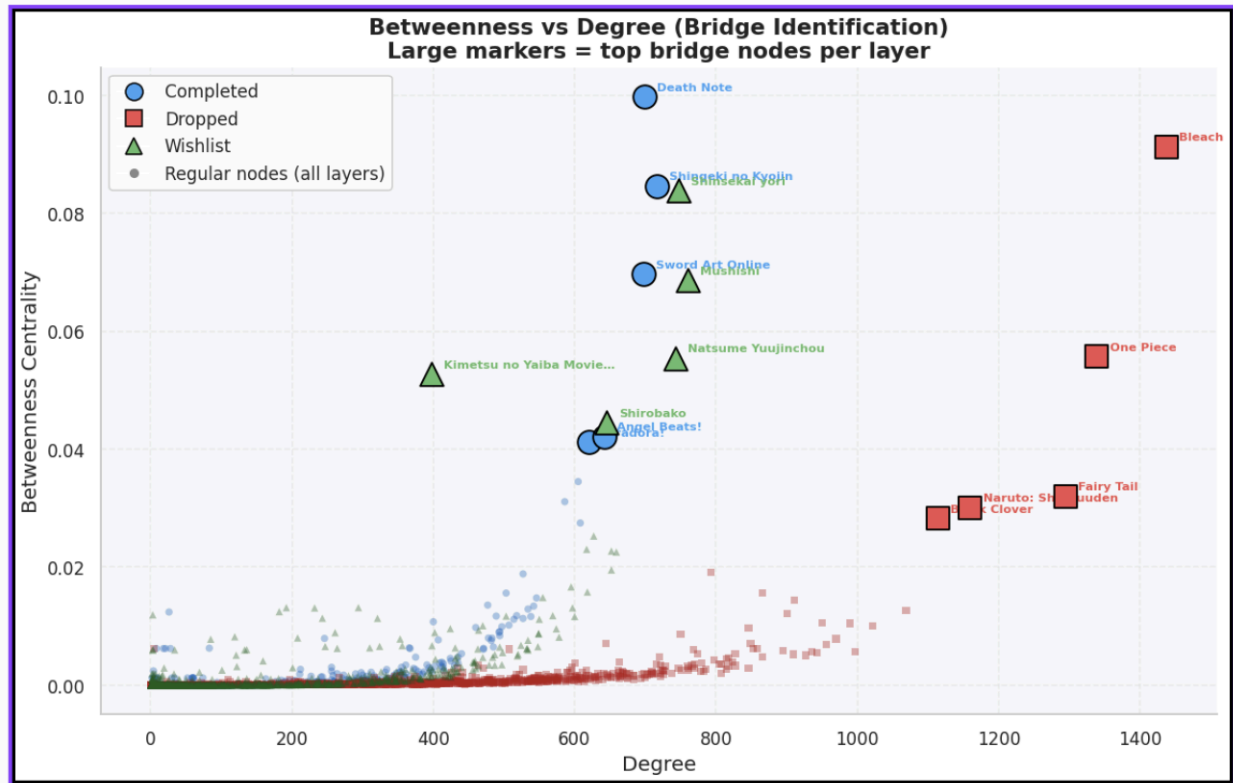
This multi-layer perspective provides a richer understanding of user behavior beyond simple co-watching.



7. Bridge Anime Analysis

We compare:

- Degree centrality (popularity)
- Betweenness centrality (bridge role)



TOP BRIDGE NODES:

Completed:

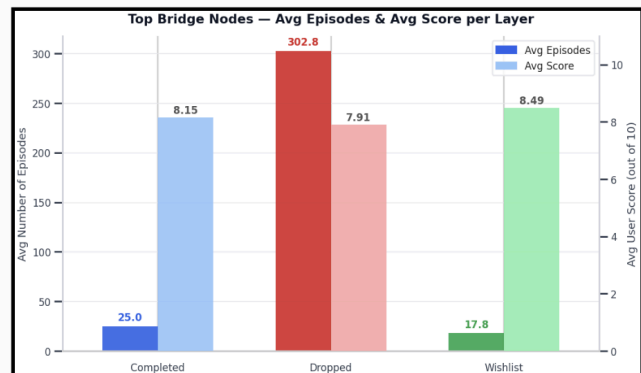
- Death Note (37 eps, score: 8.63)
- Shingeki no Kyojin (25 eps, score: 8.48)
- Sword Art Online (25 eps, score: 7.25)
- Angel Beats! (13 eps, score: 8.15)
- Toradora! (25 eps, score: 8.24)

Dropped:

- Bleach (366 eps, score: 7.8)
- One Piece (eps: N/A, score: 8.52)
- Fairy Tail (175 eps, score: 7.68)
- Naruto: Shippuuden (500 eps, score: 8.16)
- Black Clover (170 eps, score: 7.38)

Wishlist:

- Shinsekai yori (25 eps, score: 8.35)
- Mushishi (26 eps, score: 8.69)
- Natsume Yuujinchou (13 eps, score: 8.33)
- Kimetsu no Yaiba Movie: Mugen Ressha-hen (1 eps, score: 8.72)
- Shirobako (24 eps, score: 8.35)



Bridge anime connect different viewer communities across layers, completed titles link viewing patterns, dropped ones reflect shared abandonment across groups, and wishlist titles indicate cross-community interest.

Findings:

- High-degree anime are popular
- High-betweenness anime connect communities

Interpretation:

Bridge anime connect otherwise separate viewer communities, enabling interaction between different clusters of users.

- Completed layer bridges link common viewing patterns
- Dropped layer bridges reveal shared abandonment trends
- Wishlist bridges highlight cross-community interest

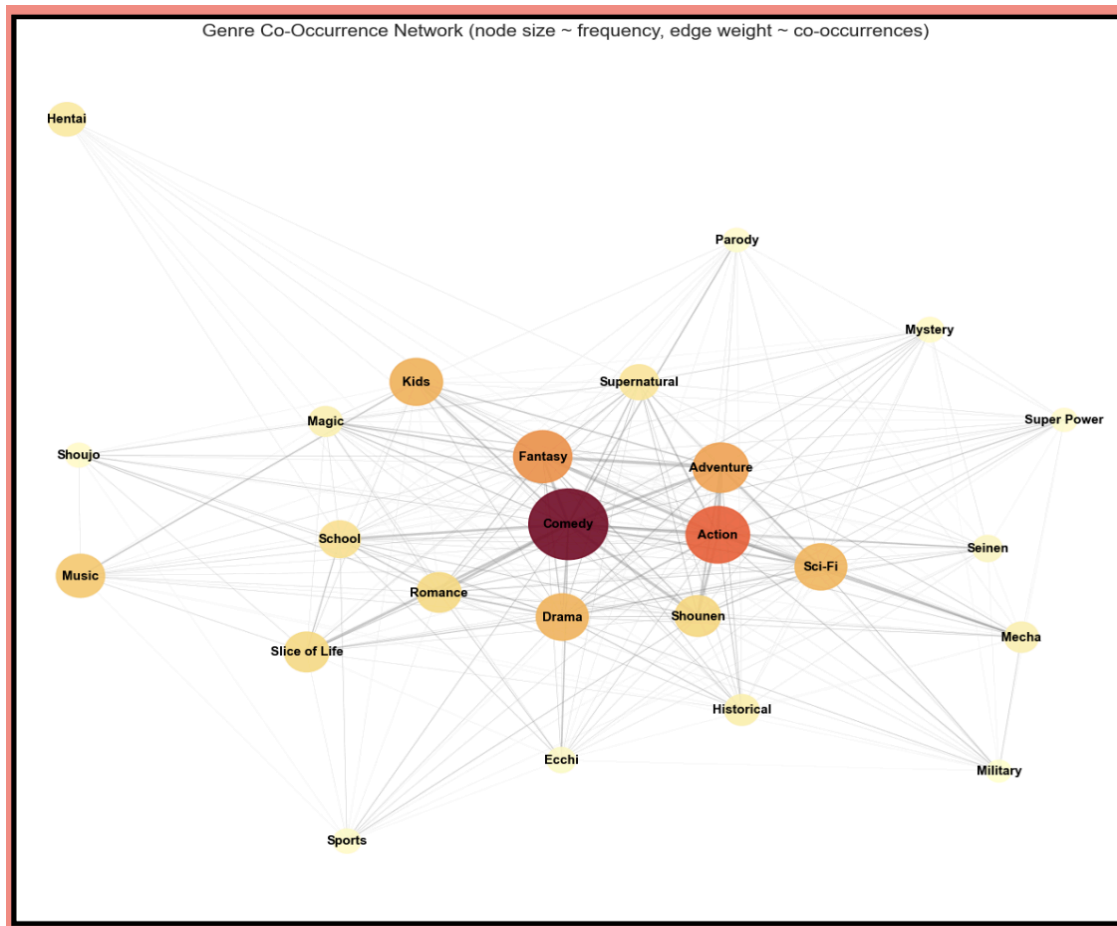
Conclusion:

- Not all popular anime are structurally critical
- Long anime are more likely to be dropped, likely due to higher time commitment
- Short, high-rated anime are more likely to be added to wishlists
- Completed anime tend to balance manageable length with perceived quality

8. Genre Network Analysis

8.1 Raw Co-occurrence

- Comedy dominates
- Network is dense but biased

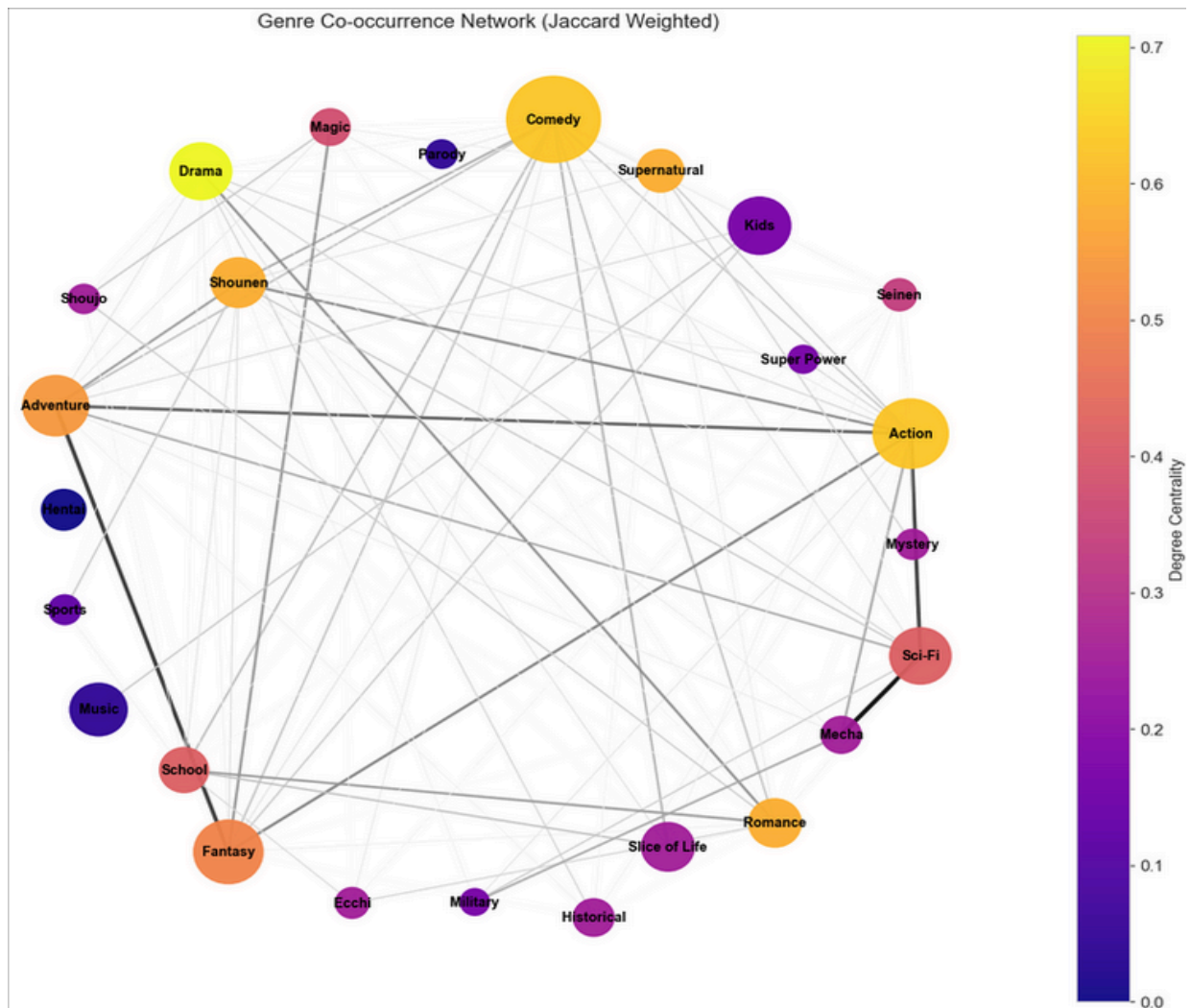


The network shows that Comedy is the central hub, connecting with many genres and appearing across diverse anime. A strong core cluster of Action, Fantasy, and Adventure forms common mainstream combinations. Overall, the network is dense, indicating high overlap between genres, while Hentai, Sports, and Music lie on the periphery, representing more niche or less frequently combined genres.

8.2 Jaccard Similarity

- Removes popularity bias
- Reveals clear clusters:
 - Action–SciFi–Mecha
 - Drama–Romance–School
 - Fantasy–Adventure

This Jaccard-weighted genre network highlights true genre similarity by removing popularity bias, resulting in a more balanced structure where no single genre dominates. Clear clusters emerge, such as Action with Sci-Fi and Mecha for tech-driven themes, Drama with Romance and School for daily-life narratives, and Fantasy with Adventure for exploration-based stories. The network is cleaner and more interpretable, making natural groupings and meaningful relationships between genres easier to identify.



Conclusion: After normalization, true genre relationships emerge more clearly, showing that raw co-occurrence alone can be misleading due to popularity effects.

9. Temporal Evolution

9.1 Network Growth

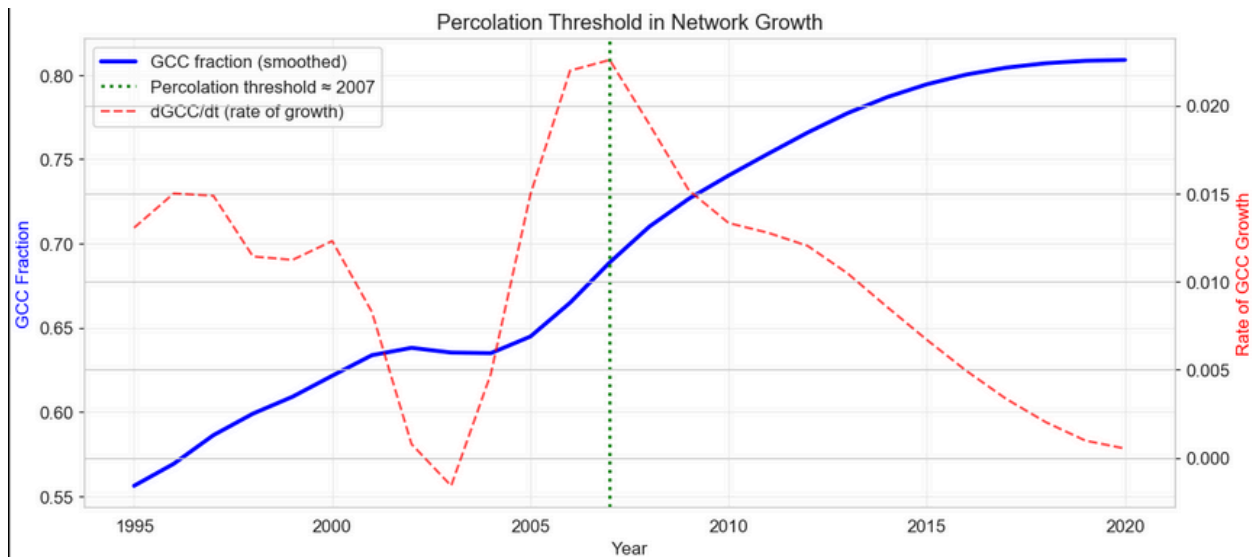
- Nodes increase steadily
- Edges grow faster → densification

9.2 Giant Component Growth

- Increasing GCC fraction over time

9.3 Percolation Transition (~2007)

- Rapid merging of clusters
- Network becomes fully connected



Conclusion:

The anime ecosystem evolved from a fragmented structure into a unified network, with a clear percolation transition marking large-scale connectivity.

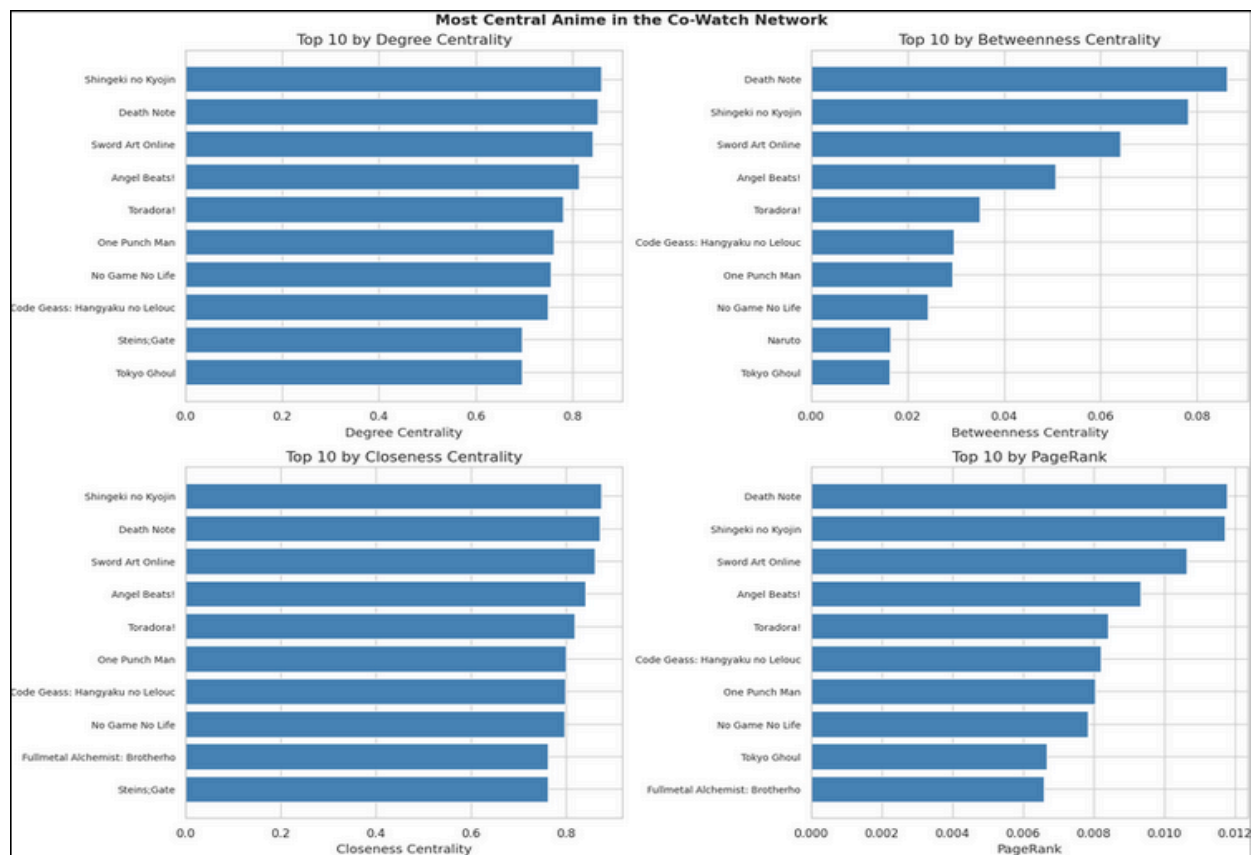
10. Centrality Analysis

We use four measures:

- Degree Centrality → popularity
- Betweenness Centrality → bridge importance
- Closeness Centrality → accessibility
- PageRank → global influence

Key findings: Anime such as *Attack on Titan*, *Death Note*, and *Sword Art Online* rank highly across all metrics.

Conclusion: These anime dominate both locally (within clusters) and globally (across the network), making them key drivers of connectivity and influence.



11. Cut Vertex Analysis

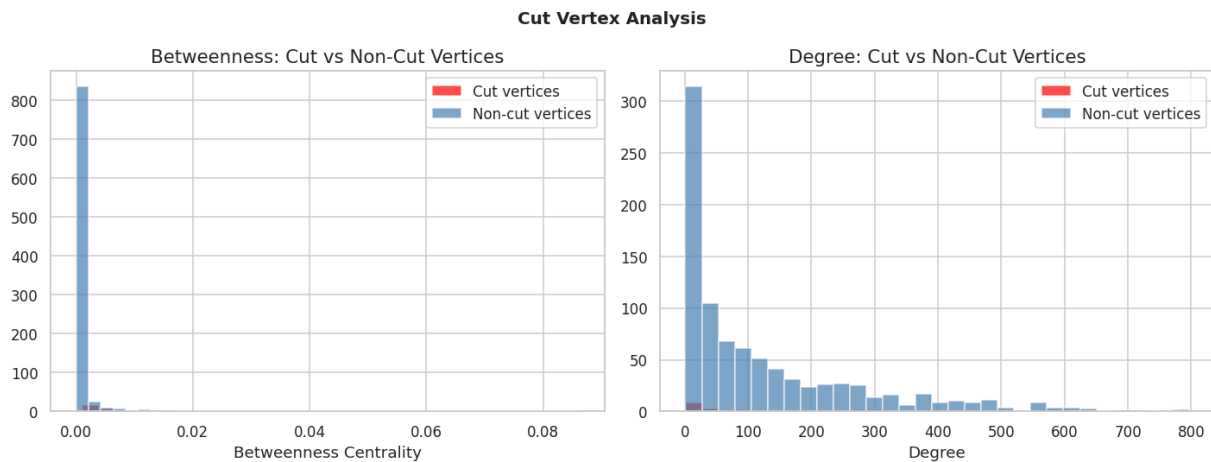
Cut vertices are nodes whose removal disconnects the network.

Findings:

- Only ~3–4% nodes are cut vertices
- These nodes:
 - Have a higher degree
 - Have a higher betweenness

Conclusion:

- Structurally critical anime are both highly connected and essential for maintaining network integrity.



12. Network Robustness

We simulate:

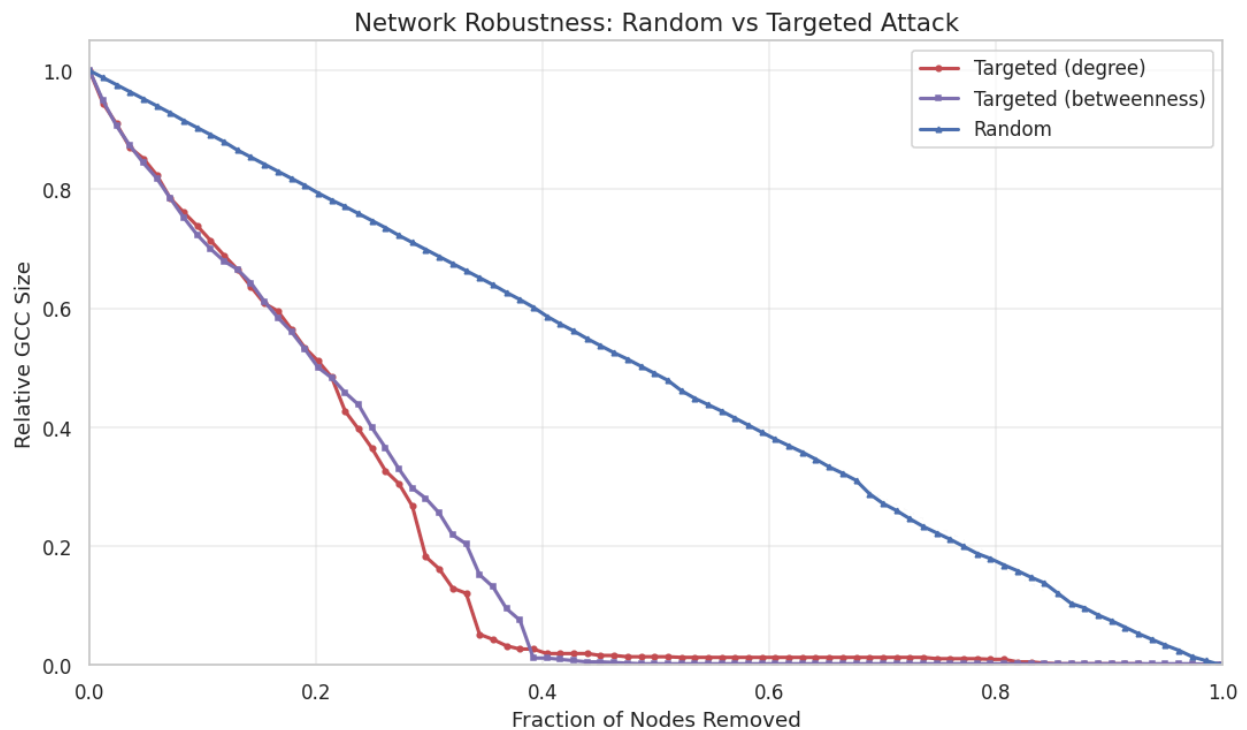
- Random removal
- Targeted removal (degree, betweenness)

Results:

- Random removal → gradual degradation
- Targeted removal → rapid collapse

Key observation:

- GCC drops sharply when top hubs are removed (~20–40%)



Conclusion: The network is robust to random failures but highly vulnerable to targeted attacks, confirming the importance of hub nodes.

13. Key Insights

- The network is hub-driven
- A small number of anime dominate connectivity
- The structure is small-world and highly clustered
- User behaviour directly shapes network formation
- The network evolves toward increasing connectivity over time

14. Limitations

- Bias toward popular anime due to thresholding
- Sampling may remove rare patterns
- Co-watch definition may ignore weak relationships

15. Conclusion

The anime viewing network is highly structured and governed by a small set of influential anime. These hubs connect diverse user communities and play a central role in shaping overall connectivity. While the system is resilient to random disruptions, it is highly dependent on key nodes, reinforcing its hub-centric nature.

This analysis highlights how user behaviour, not just content metadata, can effectively model similarity, influence, and structure in large-scale media ecosystems, with direct implications for recommendation systems and content discovery.

References

- MyAnimeList Dataset (Kaggle)
- NetworkX Documentation
- Relevant network science literature